

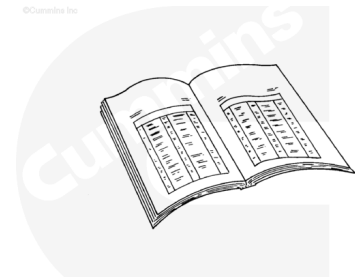
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery first and connect the negative (-) battery last.

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⚠ WARNING ⚠

Coolant is toxic; Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

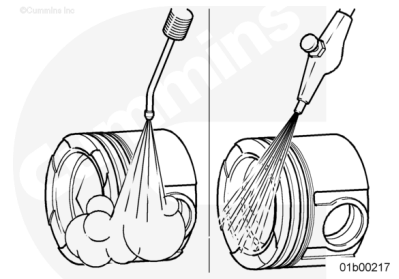
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Disconnect the battery or supply air to the starter.
- Drain the coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/89/89-008-018.html>).
- Drain the lubricating oil and remove the oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Remove the oil suction tube and block stiffener plate. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Remove the intake manifold. Refer to Procedure 010-023 (</qs3/pubsys2/xml/en/procedures/89/89-010-023.html>).
- Remove the exhaust manifold. Refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/89/89-011-007.html>).
- Remove the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>).
- Remove the rocker lever assembly. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>).
- Remove the push rods and crossheads.
- Remove the rocker lever housing. Refer to Procedure 003-013 (</qs3/pubsys2/xml/en/procedures/89/89-003-013.html>).
- Remove the cylinder head. Refer to Procedure 002-004 (</qs3/pubsys2/xml/en/procedures/89/89-002-004.html>).
- Remove the piston cooling nozzles. Refer to Procedure 001-046 (</qs3/pubsys2/xml/en/procedures/89/89-001-046.html>).
- Remove the piston and connecting rod assembly and disassemble. Refer to Procedure 001-054 (</qs3/pubsys2/xml/en/procedures/89/89-001-054.html>).

Clean and Inspect for Reuse



When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



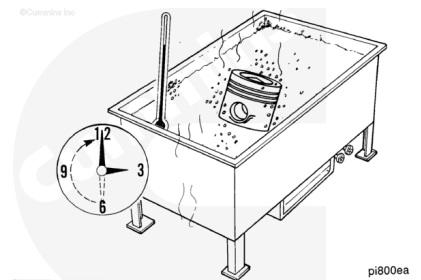
Use steam to clean the outer layers of carbon from the piston surface. The bead-blast method can be used to clean the top of the piston. If this method is used, cover the ring grooves.

The oil gallery holes **must** be covered before using the bead blast method.

It is **only** necessary to remove the carbon buildup. It is **not** necessary to make the piston crown appear new.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

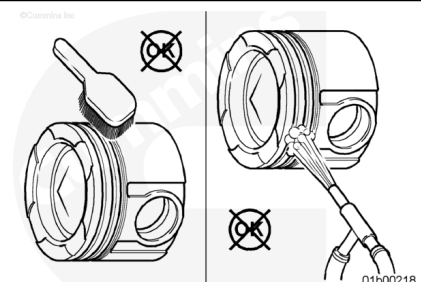
Do not contaminate the wash tank solution with bead-blast materials. Use a kerosene emulsion-based solvent. Do not use a solvent that is higher than 9.5 pH. Do not use a solvent that contains chlorinated hydrocarbons with cresol, phenols, or cresylic components. Engine damage can result.

For best results, soak the pistons several hours or overnight. Use a solvent that can be heated from 90 to 95°C [194 to 203°F]. Use a cleaning tank that will circulate and filter the solvent.

Allow the pistons to soak in solvent for a minimum of 30 minutes.

⚠ CAUTION ⚠

Do not use the bead-blast method to clean the skirt. The piston skirt will be damaged by the bead-blast method.



⚠ CAUTION ⚠

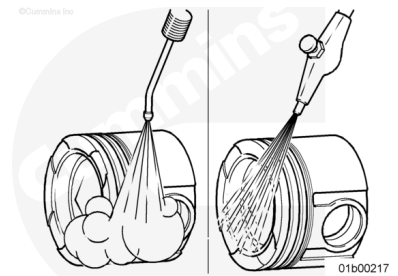
Do not use a metal bristle brush to clean the pistons. Damage to the ring grooves will result.

Use a brush to clean the ring grooves. Walnut shell blasting can be used on the piston crown.

Repeat the soaking and scrubbing process until the piston is thoroughly cleaned.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

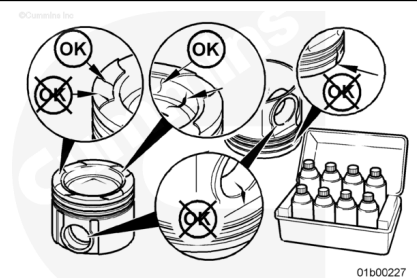
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam to rinse the solvent from the pistons.

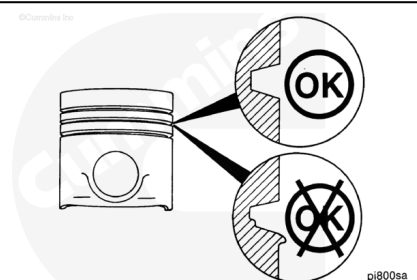
Dry the pistons with compressed air.

Use crack detection kit, Part Number 3375432, or equivalent, to inspect the bowl, crown and piston pin bore.

Remove any cover plugs in the oil gallery holes.



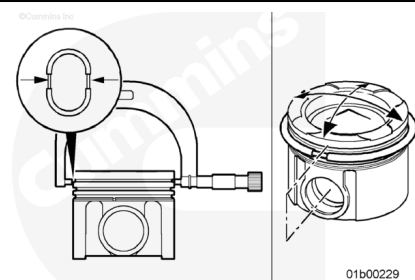
Inspect the ring grooves. The pistons **must** be replaced if there is a visible ridge in the back of the groove, or if a lip has formed on the outer diameter of the groove.



Measure

Use piston ring groove wear gauge, Part Number 3824846, and a 152- to 177-mm [6- to 7-in] micrometer to inspect the top ring groove (compression ring).

Measure the top and second ring grooves (diameter over pins) when the piston temperature is at 21°C [70°F] in two areas 90 degrees apart as illustrated.

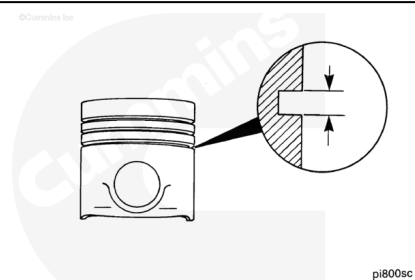


	mm		in
Top Ring	158.1	MIN	6.224
Second Ring	154.3	MIN	6.075

The piston **must** be replaced if measured dimensions are less than those in the table.

Measure the oil ring groove height.

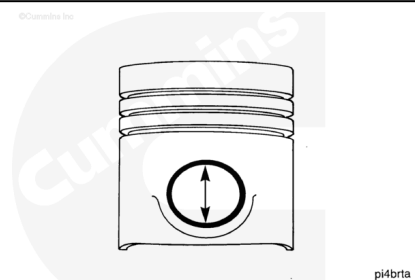
mm		in
4.01	MIN	0.158
4.03	MAX	0.159



The piston **must** be replaced if it is **not** within specifications.

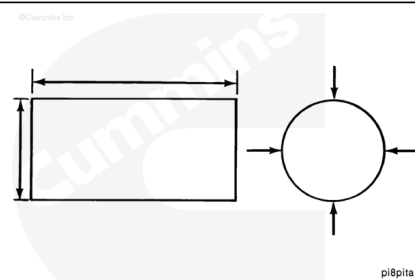
Measure the piston pin bore inside diameter.

mm		in
68.034	MIN	2.678
68.044	MAX	2.679



Measure the piston pin outside diameter and length.

mm		in
67.994	MIN	2.677
68.000	MAX	2.678



mm		in
125.80	MIN	4.953

mm		in
126.00	MAX	4.961

Finishing Steps

- Assemble and install the piston and connecting rod. Refer to Procedure 001-054 (</qs3/pubsys2/xml/en/procedures/89/89-001-054.html>).
- Install the piston cooling nozzles. Refer to Procedure 001-046 (</qs3/pubsys2/xml/en/procedures/89/89-001-046.html>).
- Install the cylinder head. Refer to Procedure 002-004 (</qs3/pubsys2/xml/en/procedures/89/89-002-004.html>).
- Install the rocker lever housing. 003-013 (</qs3/pubsys2/xml/en/procedures/89/89-003-013.html>).
- Install the push rods and crossheads.
- Install the rocker lever assembly. 003-009 (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>).
- Adjust the overhead set. Refer to Procedure 003-006 (</qs3/pubsys2/xml/en/procedures/89/89-003-006.html>).
- Install the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>).
- Install the exhaust manifold. Refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/89/89-011-007.html>).
- Install the intake manifold. Refer to Procedure 010-023 (</qs3/pubsys2/xml/en/procedures/89/89-010-023.html>).
- Install the oil suction tube and block stiffener plate. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Install the lubricating oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Fill the engine with clean 15W-40 lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>) for engine oil capacity.
- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/89/89-008-018.html>).
- Connect the battery or supply air to the starter.
- Operate the engine to normal operating temperature. Check for leaks.



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Last Modified: 30-May-2003
